Universität Konstanz

Introduction to Computing for the Social Sciences

Session 13 – Social Science Applications

David Garcia

Winter Semester 2026-2027

Course Outline

- Session 1: Introduction
- Session 2: Information Coding
- Session 3: Data Structures
- Session 4: Software
- Session 5: Algorithms
- Session 6: Recursion
- Session 7: Sorting Algorithms
- Session 8: Time Complexity
- Session 9: Theoretical Computer Science
- Session 10: Machine Learning
- Session 11: Databases
- Session 12: Parallel Programming
- Session 13: Social Science Applications
  - Thursday: Exercises & exam recap
- Session 14: Exam Review: Lectures recap

Overview

- Computational Social Science
- Digital Behavioral Data in the Social Sciences
- New Methods in the Social Sciences
- Computational Analysis of Social Science Theory
- Courses Outlook

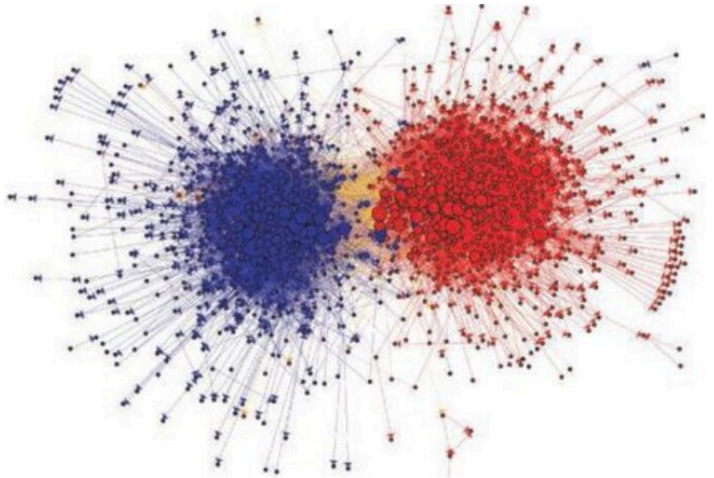
Computational Social Science (CSS)

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Computational Social Science

Quantitative testing of theories from the social sciences at unprecedented breadth, depth and scale. (Lazer et al., *Science*, 2009)

- Using new, large-scale data to inform social science theory
- Quantitative and empirical, not just theoretical
- Not data-driven studies: **interdisciplinary research**



Computational social science. D Lazer, A Pentland, L Adamic, S Aral, AL Barabási… *Science*, 2009

Buzzwords related to Computational Social Science

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Social Data Science

Focus on new methods to study human behavior and society rather than on testing or developing social science theory. Almost a synonym of CSS.

Web Science

Study of the Web as a socio-technical system. Empirical studies but also computer science methods development.

Digital Sociology

Sociology of the digital transformation and social phenomena affected by the new use of Information and Communication Technology.

Computational in Computational Social Science

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It can have three meanings, all related to computation but different:

Digital

Based on large datasets of human behavior, for example produced by the Web and social media.

Computerized

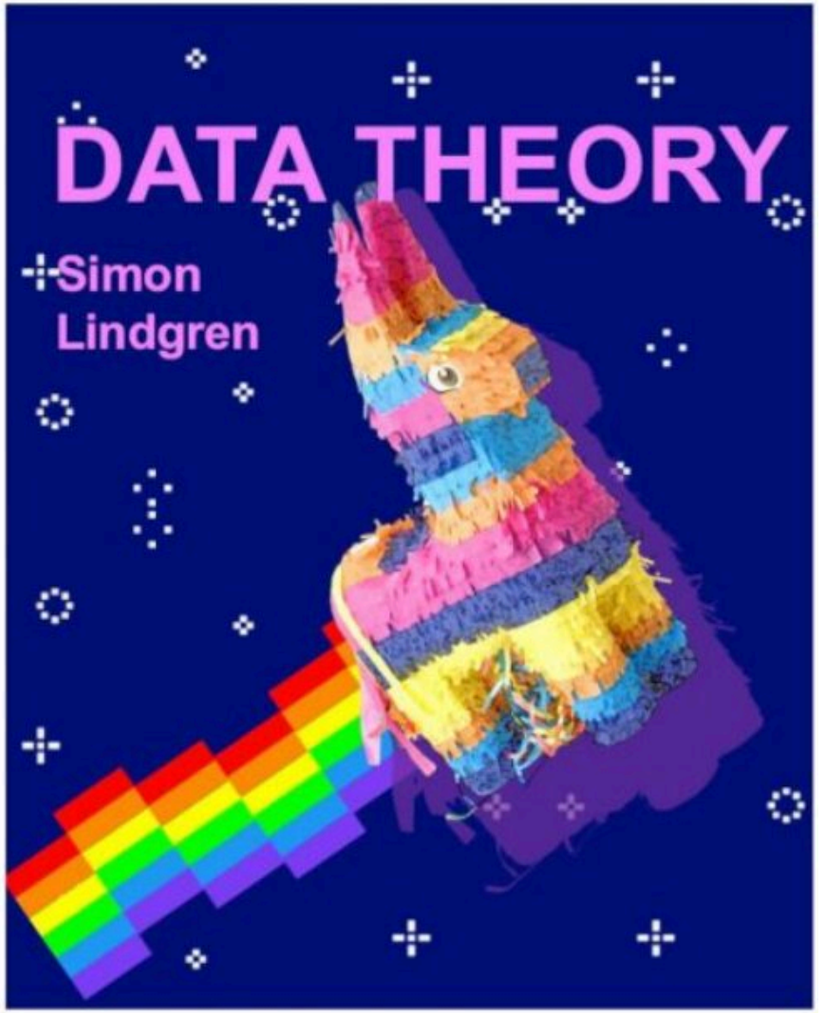
The quantitative analysis of data in an automated, tractable, repeatable and extensible fashion.

Generative

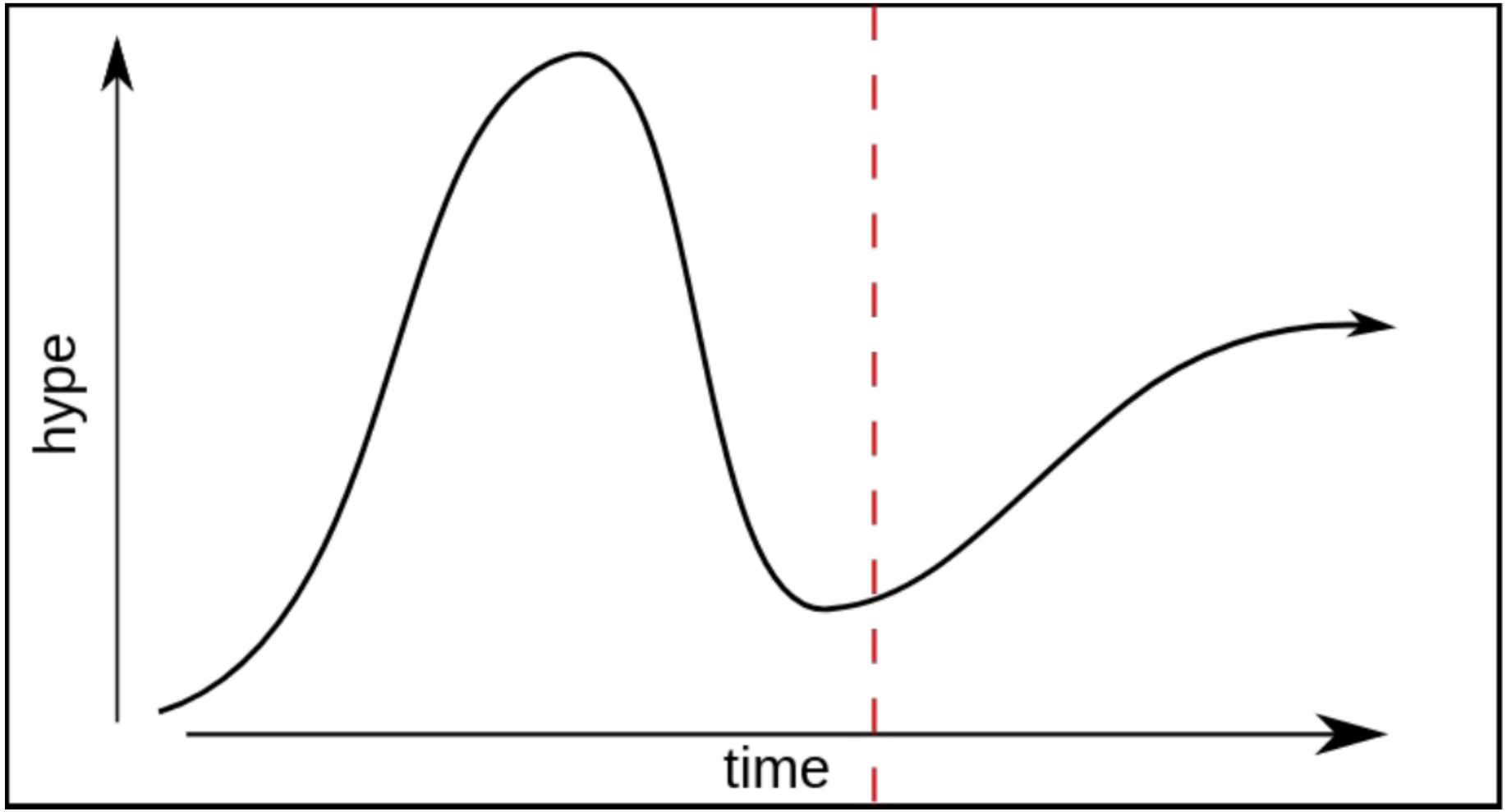
Application of data and results to the design of agent-based models that explain complex social phenomena and motivate interventions.

Early data-driven CSS



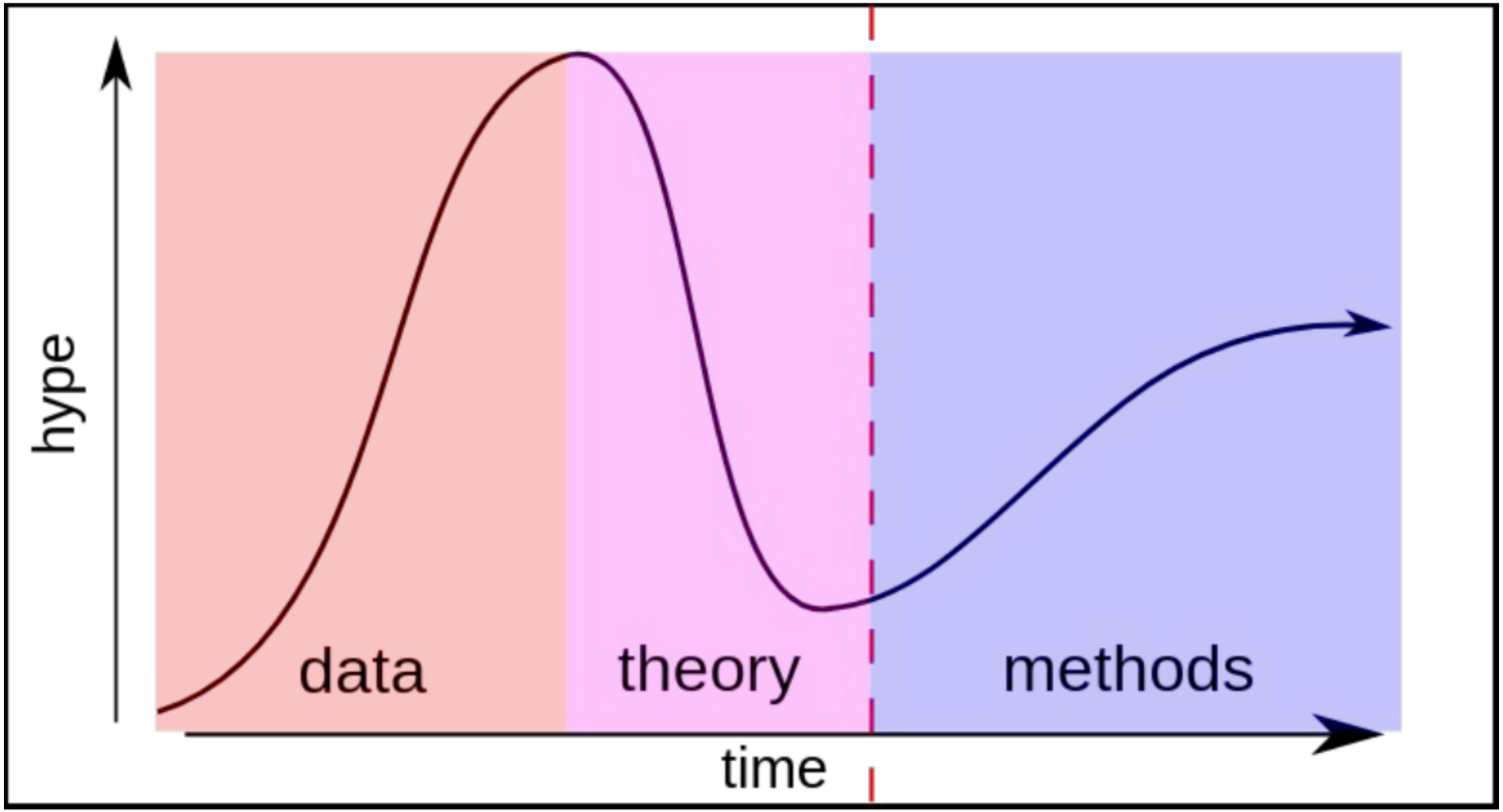


The Gartner Hype Curve of CSS

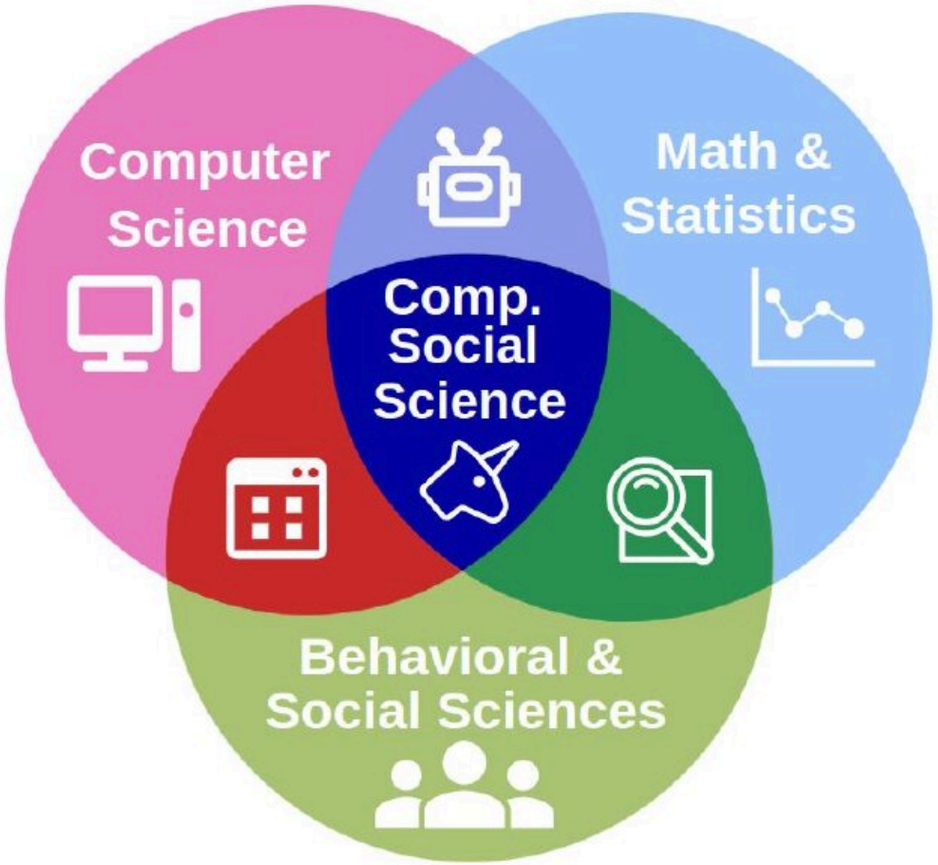


Bit by Bit: Social Research in the Digital Age. Matthew Salganik, 2017

The Three Ages of CSS

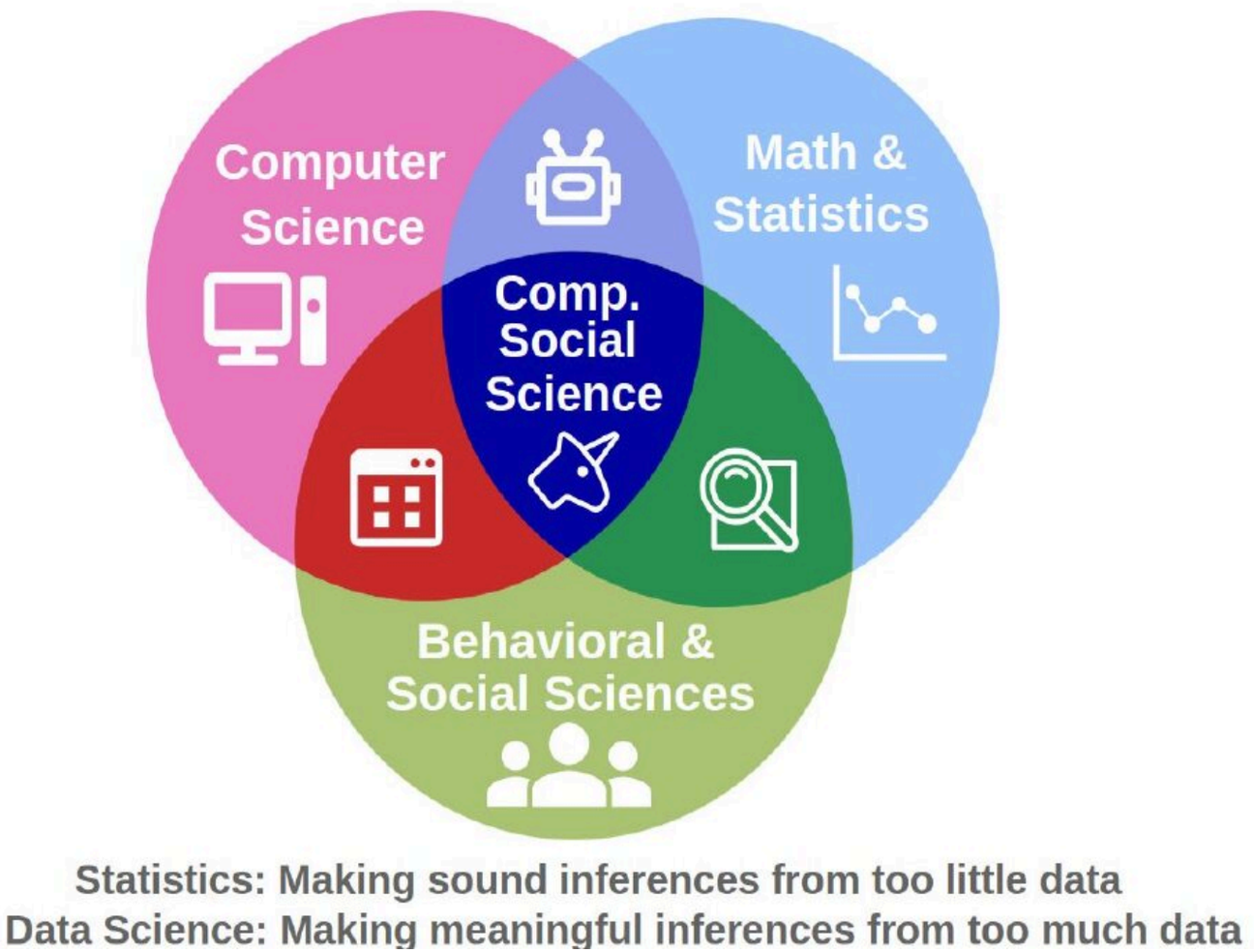


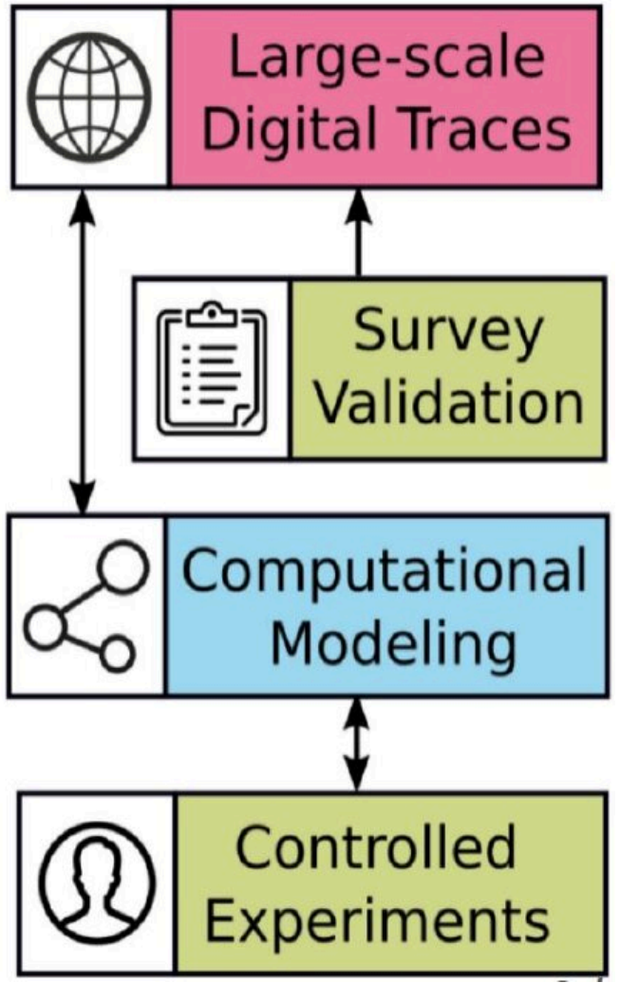
Combining Disciplines and Methods in CSS



Statistics: Making sound inferences from too little data
Data Science: Making meaningful inferences from too much data

Combining Disciplines and Methods in CSS





Digital Behavioral Data in the Social Sciences

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- Computational Social Science
- **Digital Behavioral Data in the Social Sciences**
- New Methods in the Social Sciences
- Computational Analysis of Social Science Theory
- Courses Outlook

Sensors and connectivity are increasingly pervasive



Digital Behavioral Data in the Social Sciences

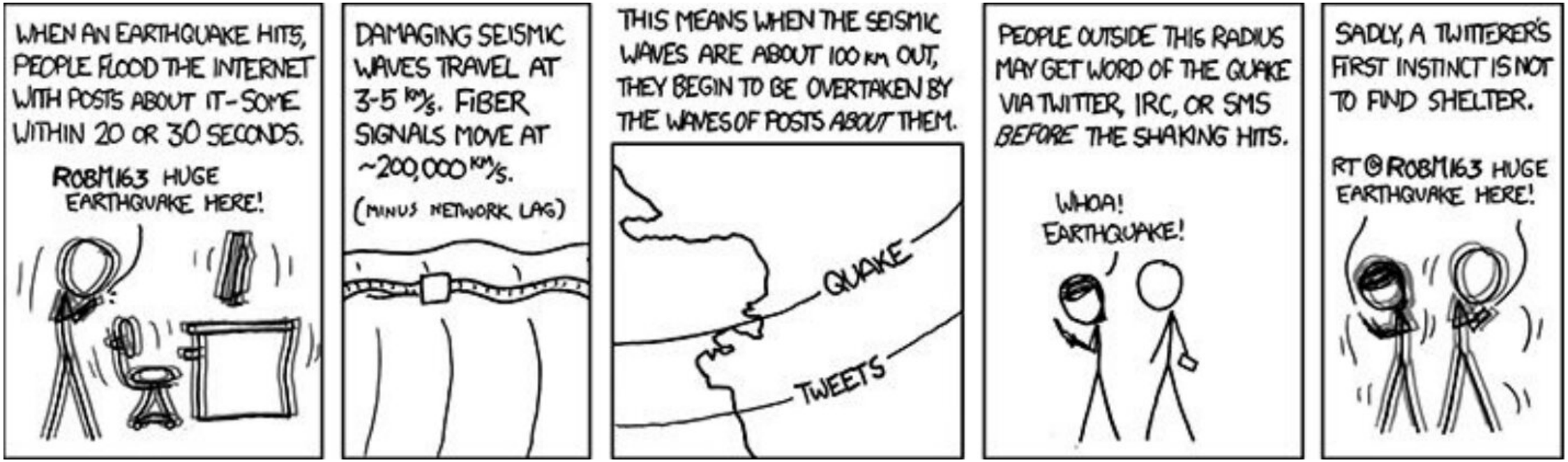
More and more of our actions happen through digital technologies, e.g.:



- Searching for a job online
- Talking with a family member over the phone
- Sharing open source code
- Posting a video on a social networking site
- Paying at the supermarket with a credit card
- Chatting on a company instant messenger

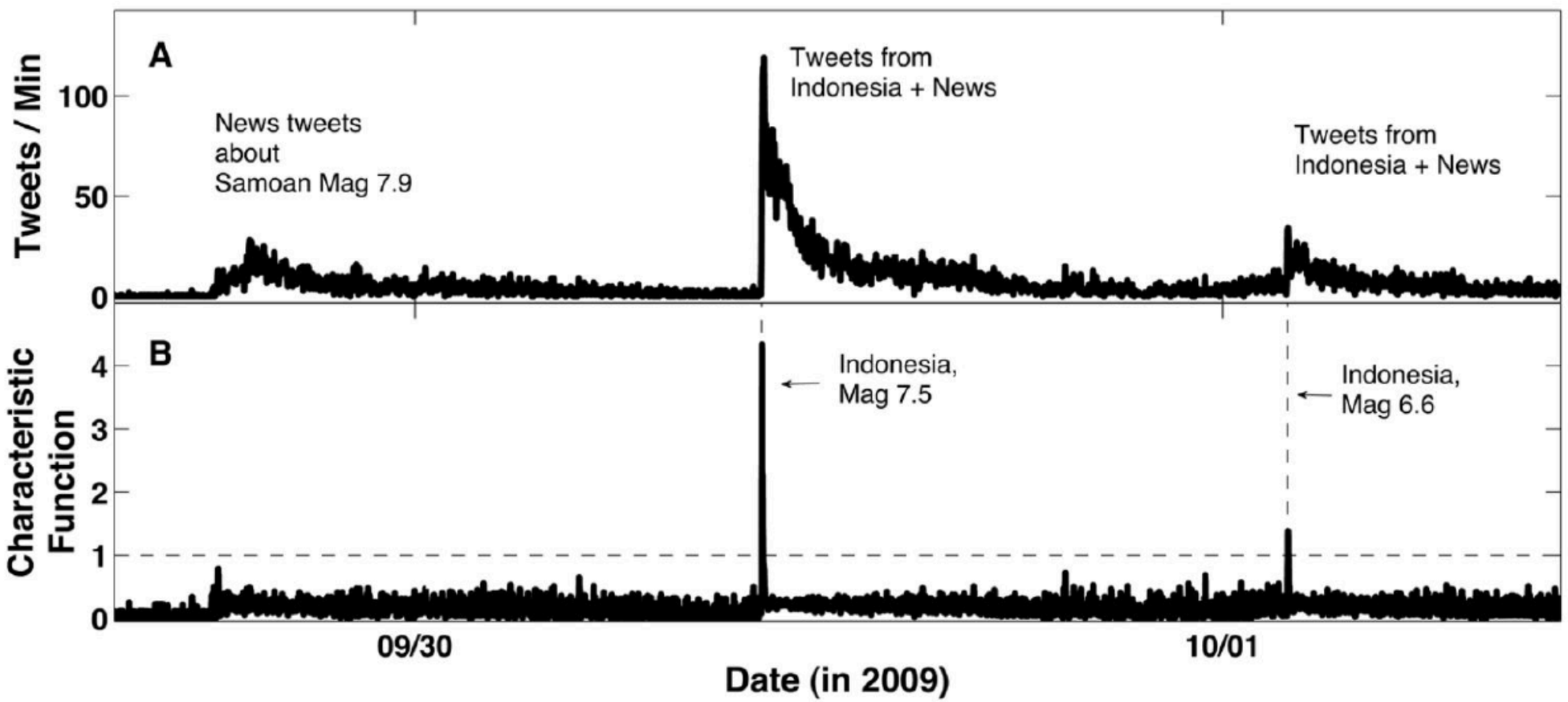
All of these actions generate **digital traces** that can be analyzed in social science research

Research example: Earthquakes on Twitter



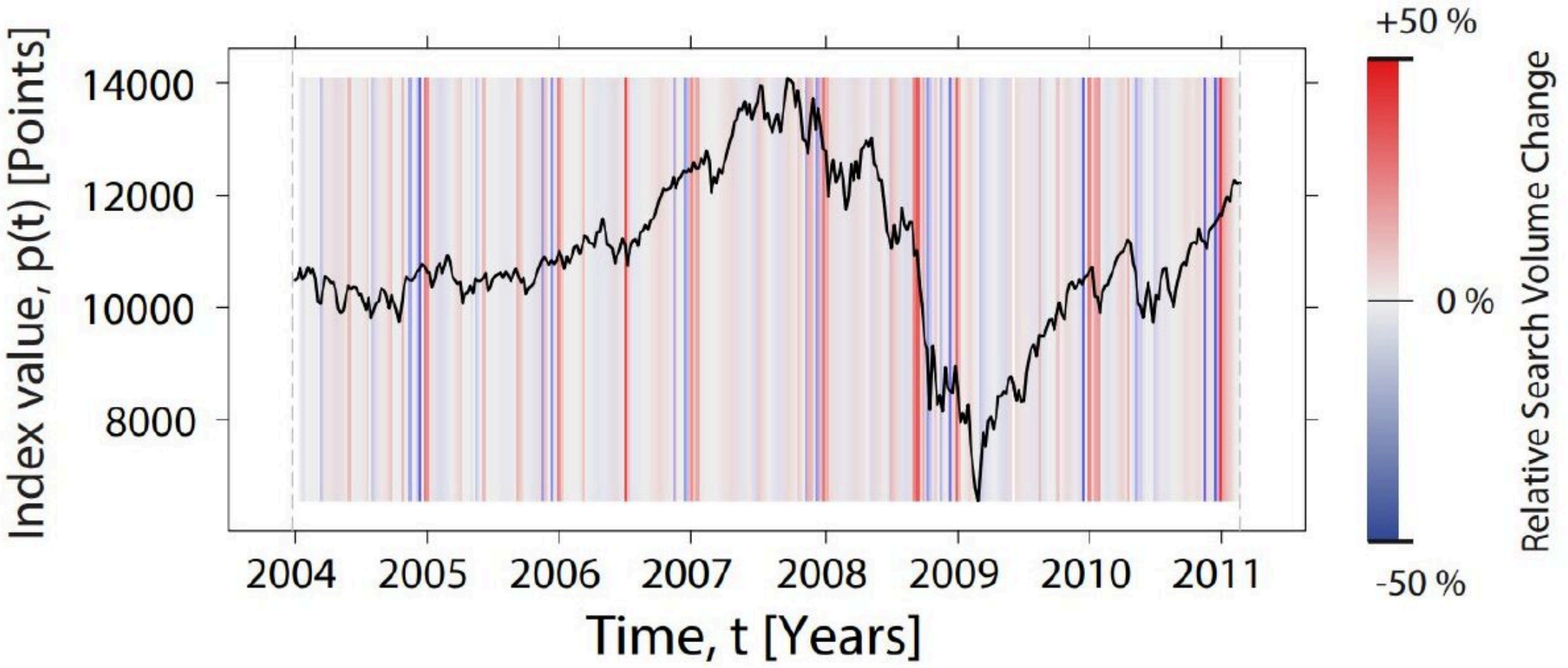
XKCD 723, *Seismic Waves*: https://xkcd.com/723/

Research example: Earthquakes on Twitter



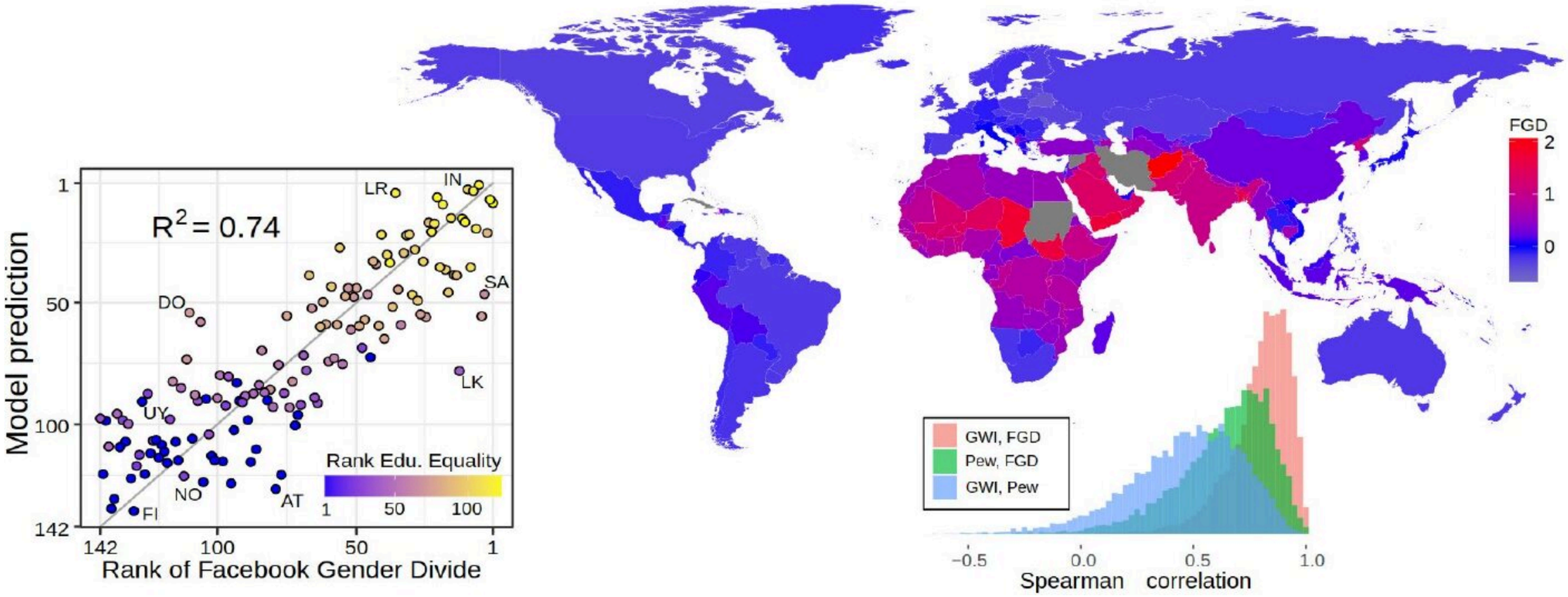
Earle PS, Bowden DC, Guy M. Twitter earthquake detection: earthquake monitoring in a social world. *Annals of Geophysics* (2012)

Research example: Trading and Google trends



Quantifying Trading Behavior in Financial Markets Using Google Trends. Tobias Preis, Helen Susannah Moat, H Eugene Stanley. *Scientific Reports* (2013)

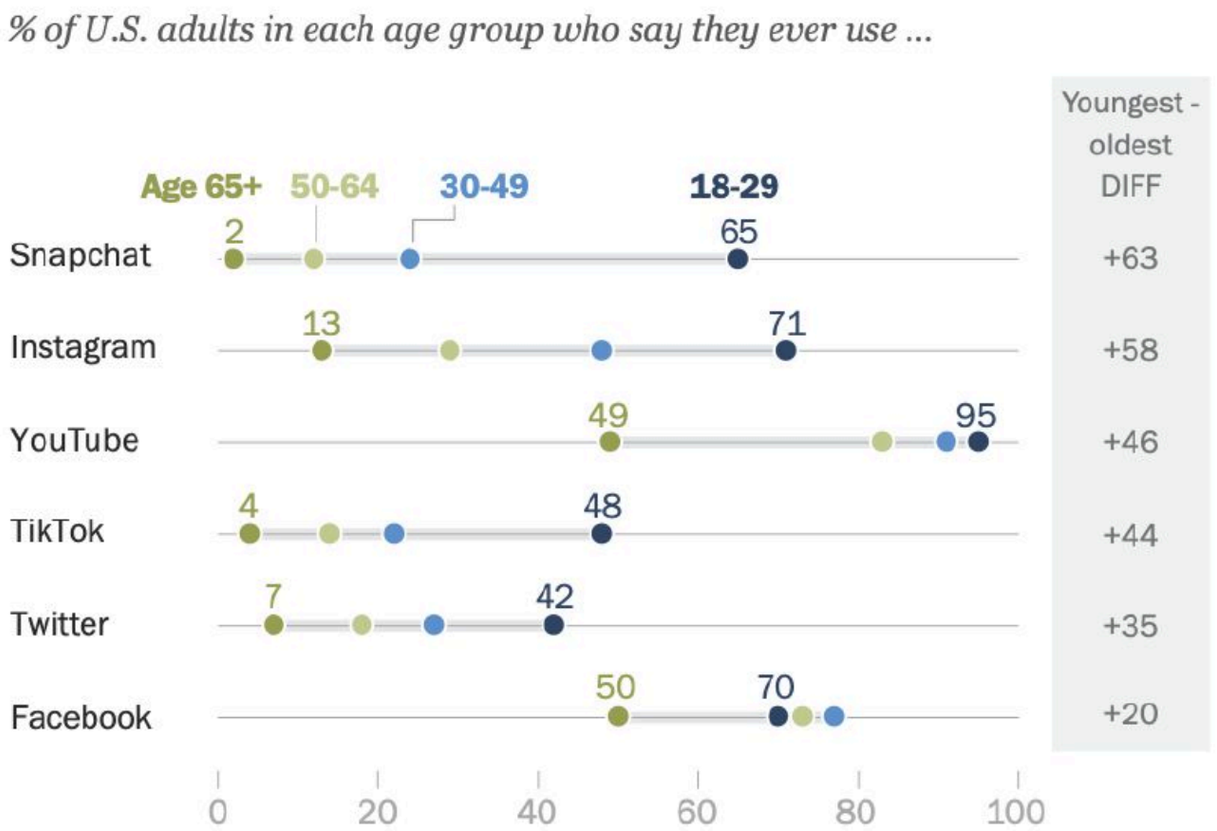
Research example: Gender Inequality with Facebook



Garcia, D., Mitike Kassa, Y., Cuevas, A., Cebrian, M., Moro, E., Rahwan, I., & Cuevas, R. Analyzing gender inequality through large-scale Facebook advertising data. *PNAS* (2018)

Issues with Digital Behavioral Data: Representation





Meaningful measures of human society in the twenty-first century. David Lazer et al., *Nature* (2021)

https://www.pewresearch.org/internet/2021/04/07/social-media-use-in-2021/

New Methods in the Social Sciences

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The Challenge of New Unstructured Data

Data sets
in tutorials



Data sets in
the wild



https://twitter.com/mszll/status/1332245497617190913

Heterogenous Data Sources and Methods

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- Social science data used to have few formats: e.g. experiment outcome annotations or survey responses
- Now many more formats and data sources:
  - text, social networks, locations and timestamps, likes…
- Sometimes these formats are even related to each other
- Methods have been improving over time:
  - **First generation of methods:** adapting previous social science methods
    - example: dictionaries in text analysis
  - **Second generation of methods:** using Machine Learning models to generalize measurement from many examples of measurements
    - example: sentiment analysis
  - **Third generation of methods:** using large models to analyze highly-dimensional data
    - example: word embeddings – see the “document analysis” course

Research Example: VADER sentiment analysis

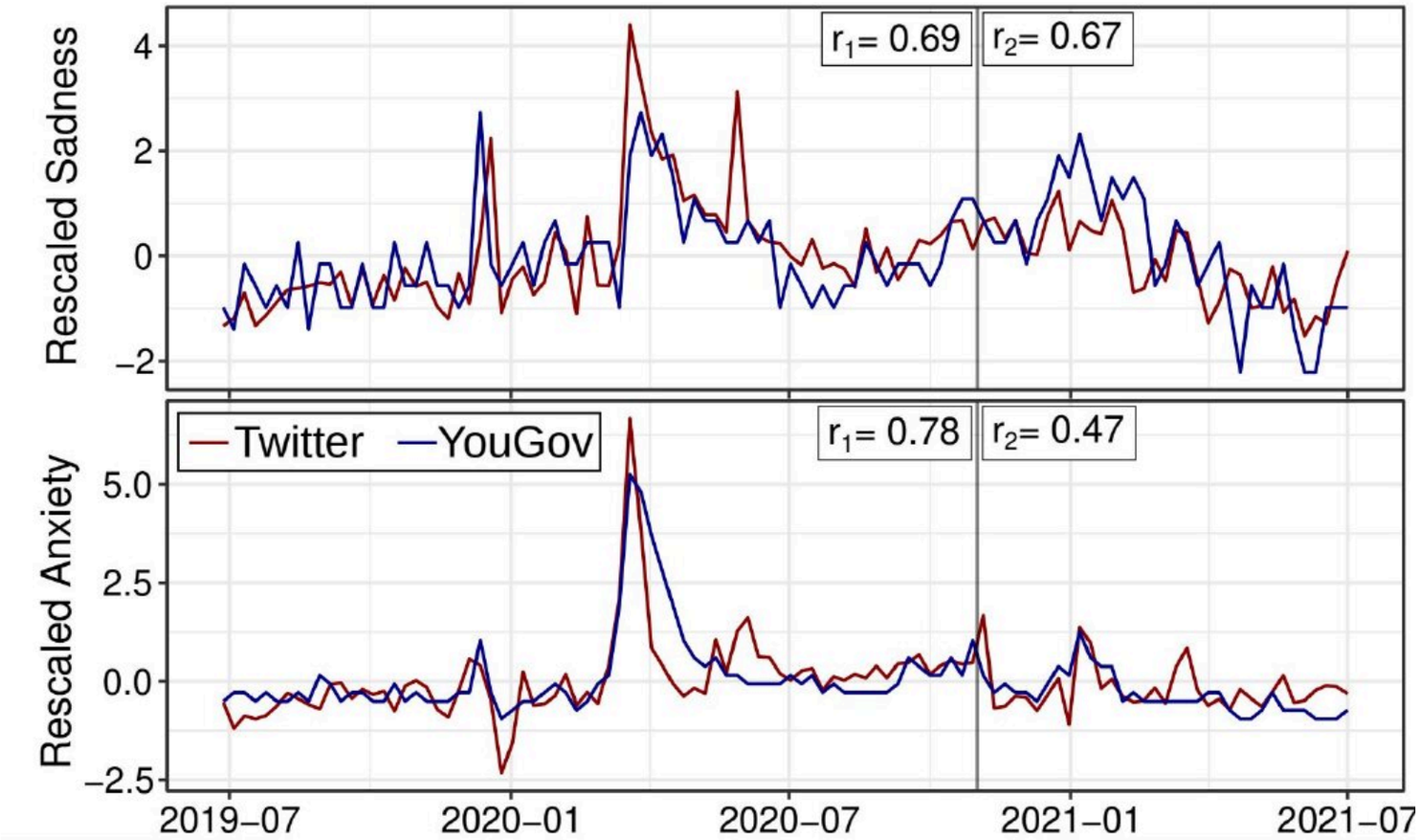


Test Condition	Example Text
Baseline	Yay. Another good phone interview.
Punctuation1	Yay! Another good phone interview!
Punctuation1 + Degree Mod.	Yay! Another extremely good phone interview!
Punctuation2	Yay!! Another good phone interview!!
Capitalization	YAY. Another GOOD phone interview.
Punct1 + Cap.	YAY! Another GOOD phone interview!
Punct2 + Cap.	YAY!! Another GOOD phone interview!!
Punct3 + Cap.	YAY!!! Another GOOD phone interview!!!
Punct3 + Cap. + Degree Mod.	YAY!!! Another EXTREMELY GOOD phone interview!!!

Table 2: Example of baseline text with eight test conditions comprised of grammatical and syntactical variations.

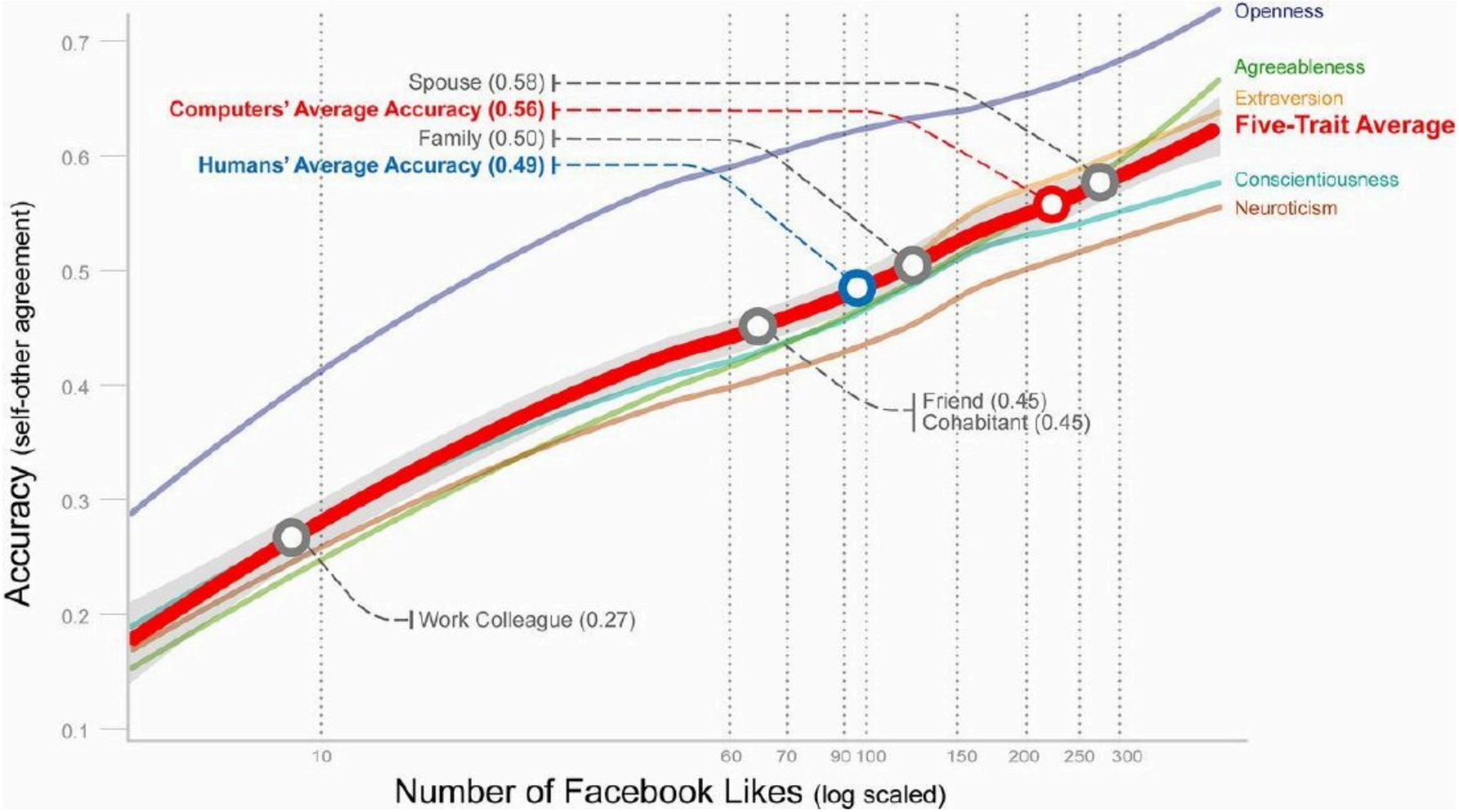
VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. C. Hutto, E. Gilbert. *ICWSM* (2014)

Research Example: Social Media and Emotion Surveys



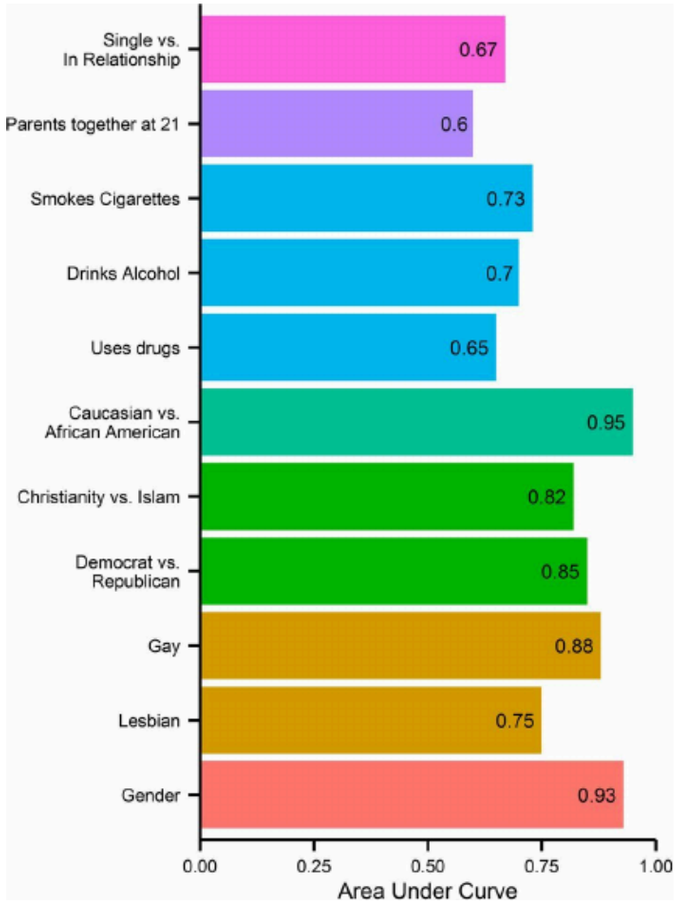
Social media emotion macroscopes reflect emotional experiences in society at large. D Garcia et al.

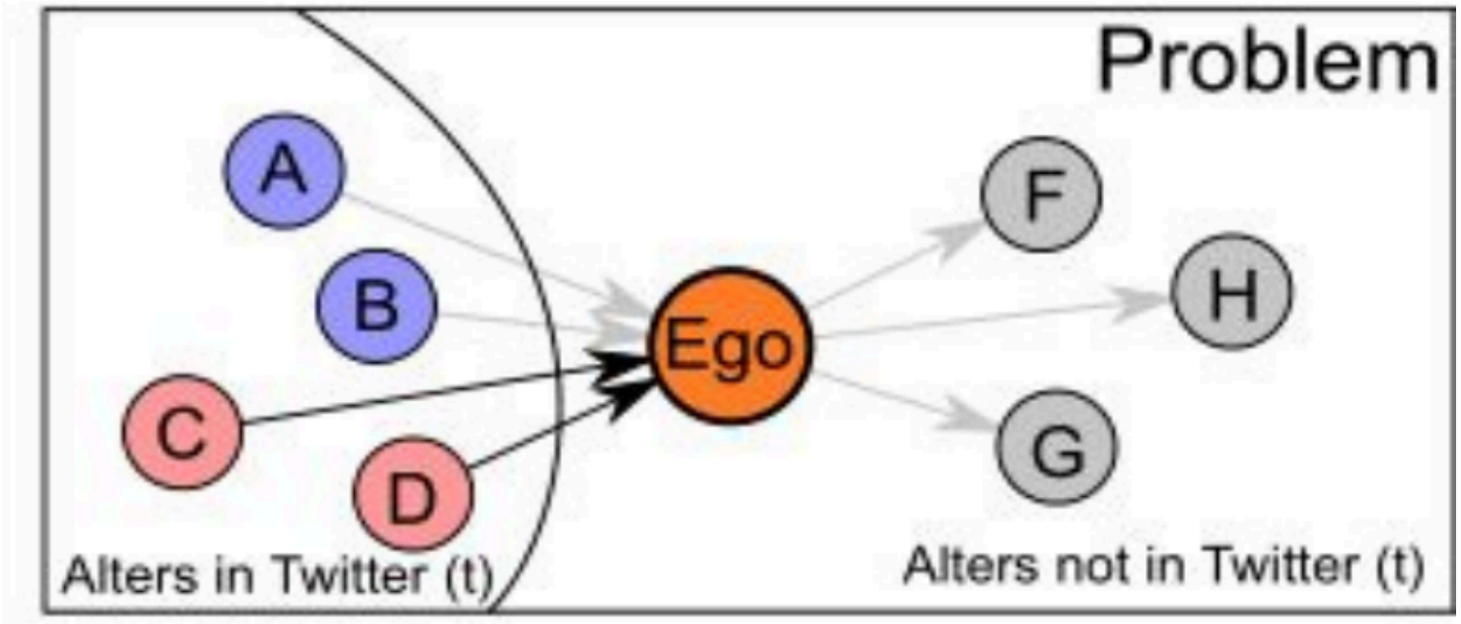
Research Example: Personality and Facebook Likes



Computer-based personality judgments are more accurate than those made by humans. Youyou et al. *PNAS* (2015)

Issues with Methods: New Privacy Risks





Private traits and attributes are predictable from digital records of human behavior. Kosinski et al. (2013)
Collective aspects of privacy in the Twitter social network. Garcia et al. (2018)

Computational Analysis of Social Science Theory

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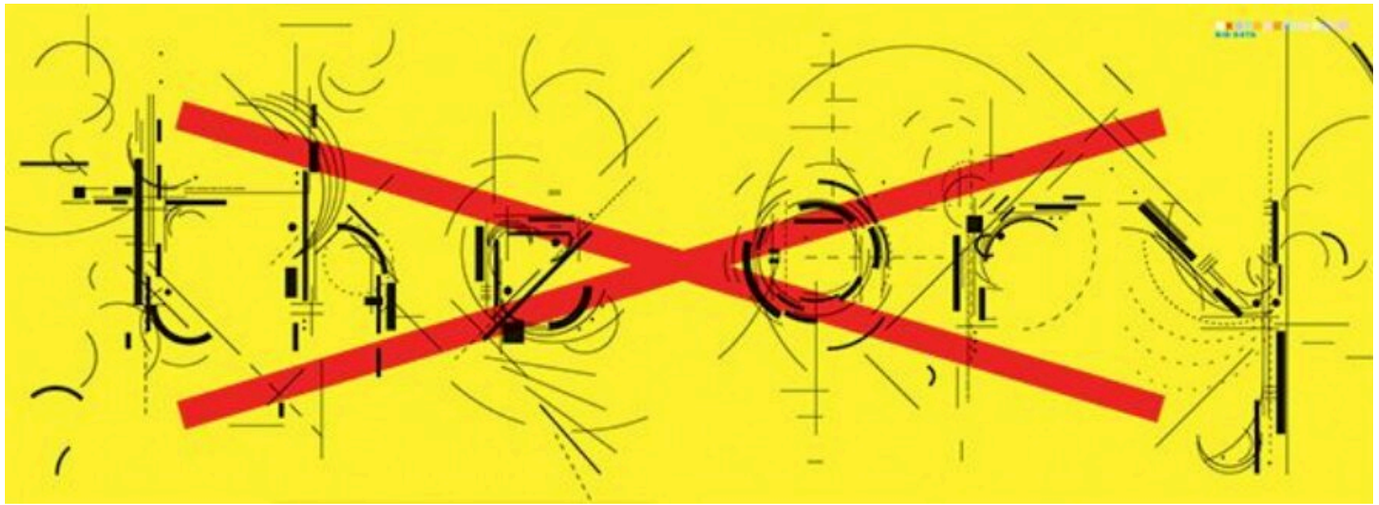
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The End of Theory?

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The End of Theory: The Data Deluge Makes the Scientific Method Obsolete

Illustration: Marian Bantjes "All models are wrong, but some are useful." So proclaimed statistician George Box 30 years ago, and he was right. But what choice did we have? Only models, from cosmological equations to theories of human behavior, seemed to be able to consistently, if imperfectly, explain the world around us. Until now. Today companies [...]



The End of Theory?

PHILOSOPHICAL TRANSACTIONS A

rsta.royalsocietypublishing.org

Opinion piece

Cite this article: Coveney PV, Dougherty ER, Highfield RR. 2016 Big data need big theory too. *Phil. Trans. R. Soc. A* **374**: 20160153. http://dx.doi.org/10.1098/rsta.2016.0153

Big data need big theory too

Peter V. Coveney[1], Edward R. Dougherty[2] and Roger R. Highfield[3]

[1]Centre for Computational Science, University College London, Gordon Street, London WC1H 0AJ, UK

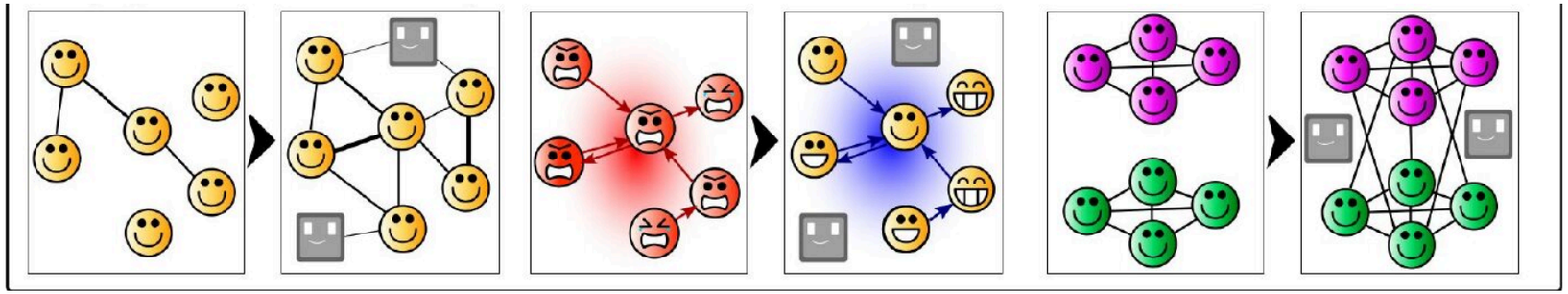
[2]Center for Bioinformatics and Genomic Systems Engineering, Texas A&M University, College Station, TX 77843-31283, USA

[3]Science Museum, Exhibition Road, London SW7 2DD, UK

PVC, 0000-0002-8787-7256

The current interest in big data, machine learning

Agent-Based Modeling: Computing and Social Theory



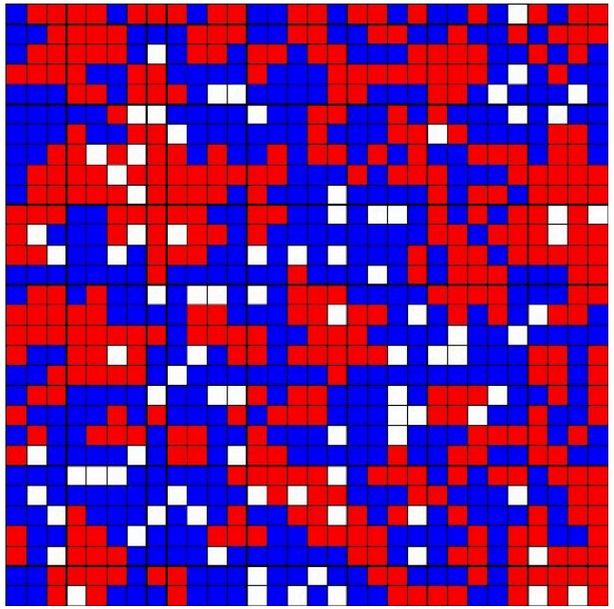
- Model humans as **agents** that symbolize a simplified behavior
- Agents have quantifiable variables and properties
  - all interaction rules are explicitly defined and parametrized
- Simulations and mathematical analysis can be applied to derive system properties that might not be easy to conclude from intuition alone
- Generates hypotheses and proofs of **generative sufficiency**, but not empirical evidence

Modeling Segregation: Schelling's Model

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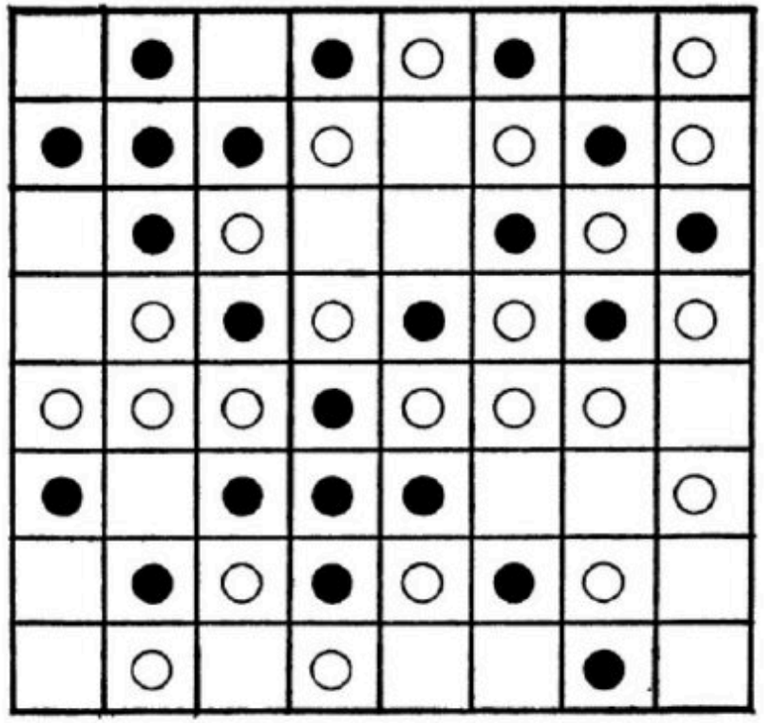
Model properties:

- Agents of two kinds, similar number
- Low but nonzero fraction of empty cells
- Cells have a neighborhood of 8 cells around the central cell
- Corners and borders have fewer neighbors
- Agents are aware of the fraction of similar agents in their neighborhood: f
- Agents are satisfied with $f \geq F$
- If unsatisfied, they jump to the closest satisfying cell
- F is a parameter that measures **intolerance**
- Example: simulation with $F = 0.2$



More on Schelling's model

Schelling's model setup



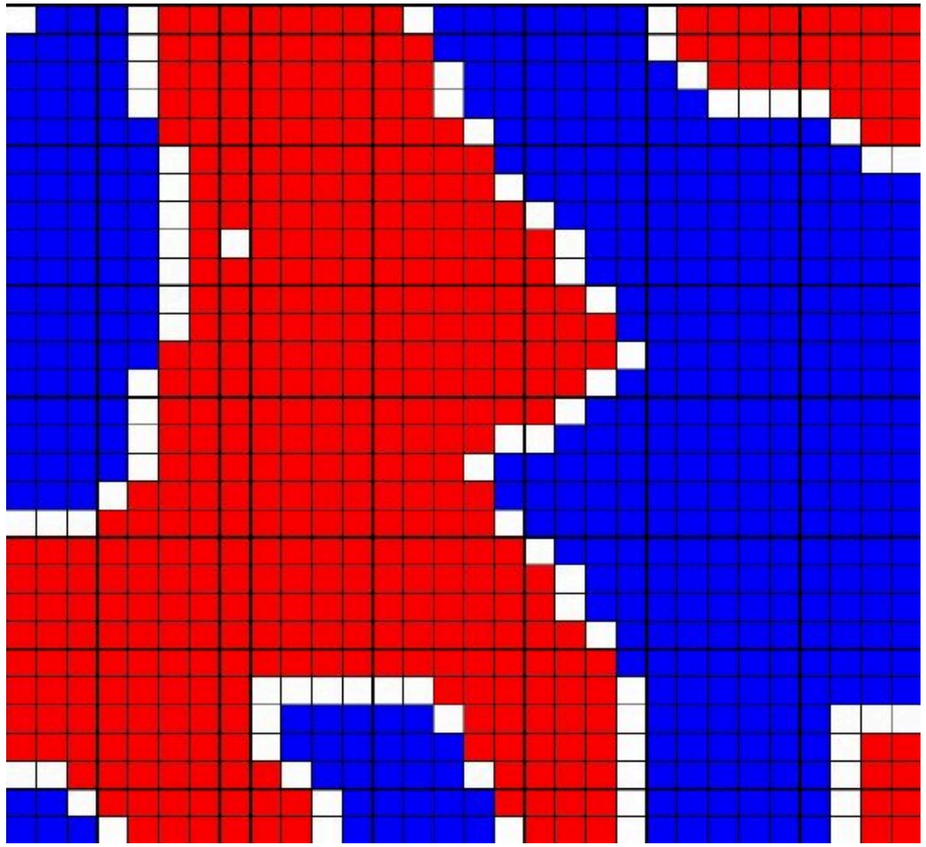
- Agents of two kinds, similar number (here 22 and 23)
- Low but nonzero fraction of empty cells (here 19)
- Cells have a neighborhood of 8 cells (3x3 - 1)
- Border and corner cells have smaller neighborhood
- Agents are aware of the fraction of similar agents in their neighborhood: f
- Agents are satisfied with $f \geq F$
- F measures intolerance

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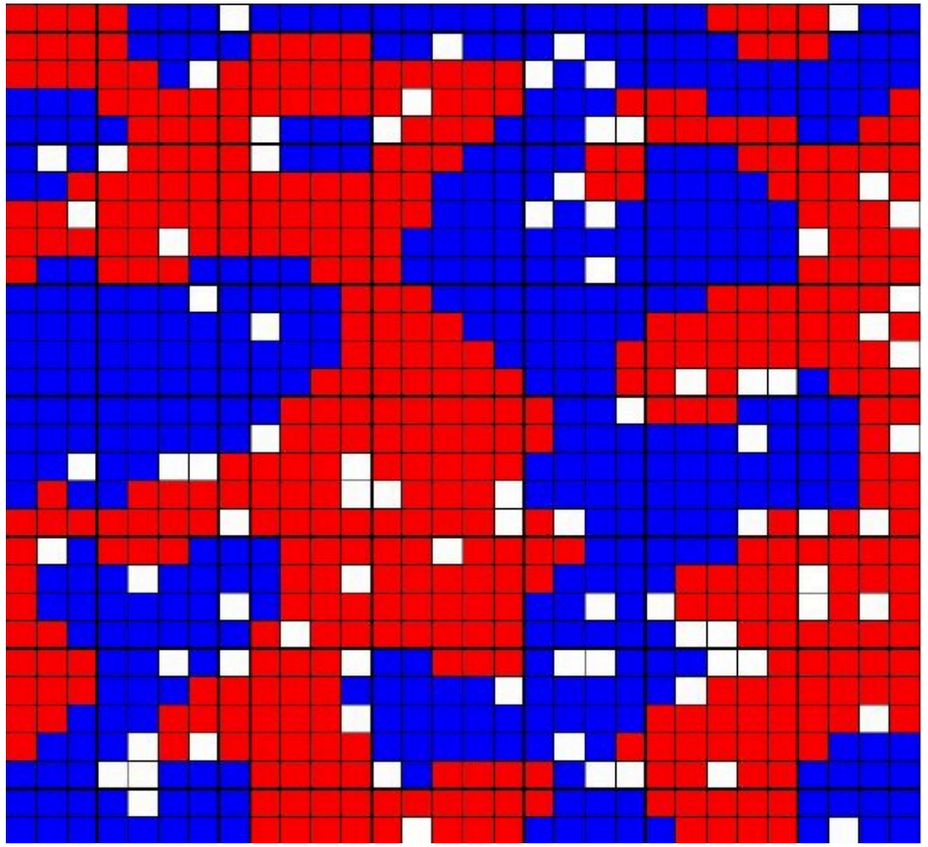
David Garcia - Computational Modelling of Social Systems

https://dgarcia-eu.github.io/CMSS-Konstanz/02_Segregation/Slides/Slides.html#14 · slides 14–38

Segregation Under Tolerance in Schelling's Model



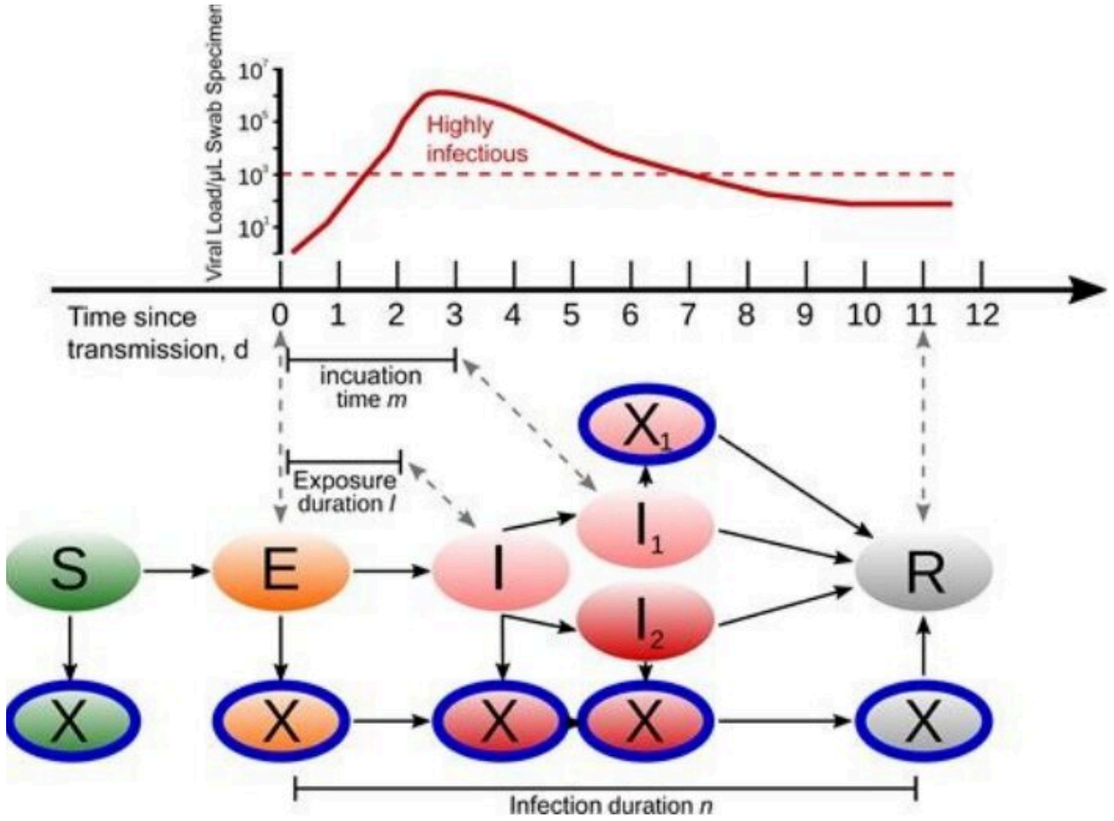
Example with $F = 0.6$. Strong segregation

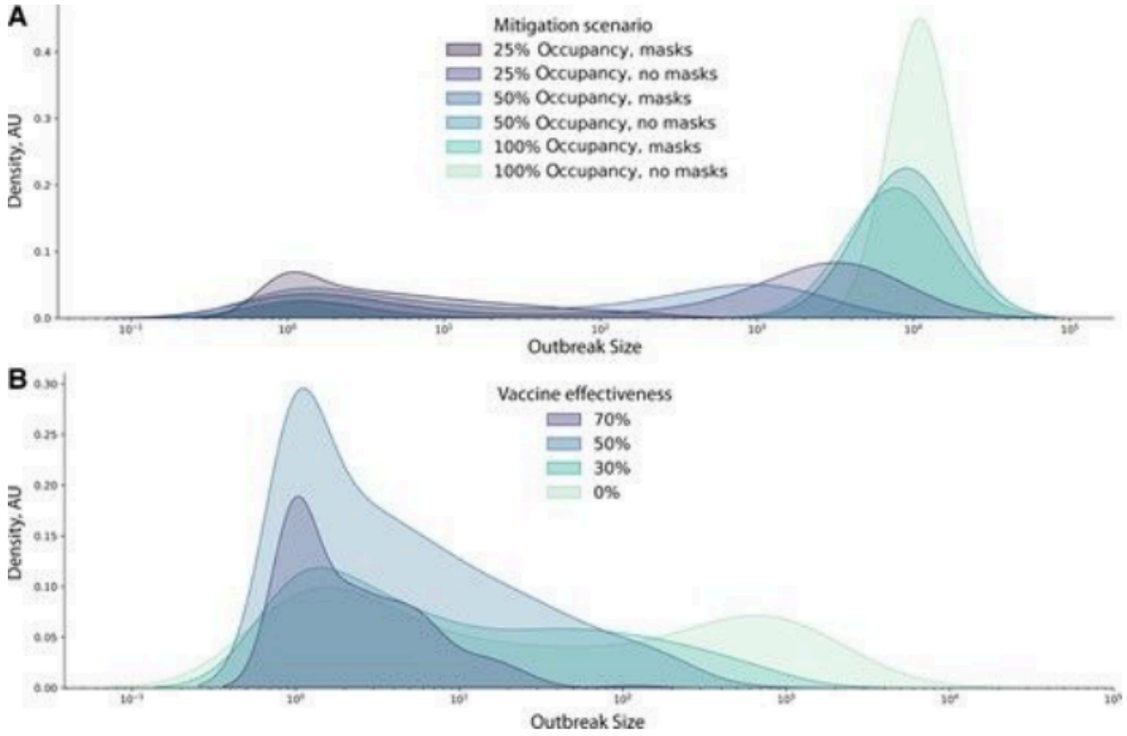


Example with $F = 0.4$. Also segregated!

Dynamic Models of Segregation. Thomas Schelling. *Journal of Mathematical Sociology* (1971)

Simulating Effects of Anti-COVID Policies





Assessment of the Effectiveness of Omicron Transmission Mitigation Strategies for European Universities Using an Agent-Based Network Model. J. Lasser, T. Hell, D. Garcia. *Clinical Infectious Diseases* (2022)

Courses Outlook

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Deep Learning for the Social Sciences

- 9 ECTS (2+2 SWS)
- Taught next SS by Giordano De Marzo
- Machine Learning concepts
- Multilayer perceptrons and their limitations
- Convolutional Neural Networks
- RNN, LSTM, RL
- Applications to time series, graphs, generative modeling and NLP
- Exercises in PyTorch



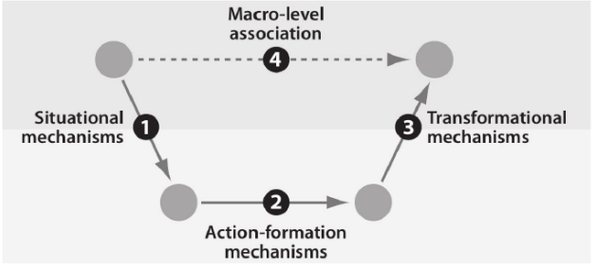
2027: Social Media Data Analysis

- 9 ECTS (2+2 SWS)
- To be taught in 2027 by David Garcia
- Course contents:
  - Analysis of search data to measure information seeking
  - Popularity and social impact in social media
  - Application and evaluation of sentiment analysis techniques for social media text
  - Methods and applications of social network analysis in online media
- Practical assignments where you get and analyze your own data
- https://github.com/dgarcia-eu/SocialMediaDataAnalysis



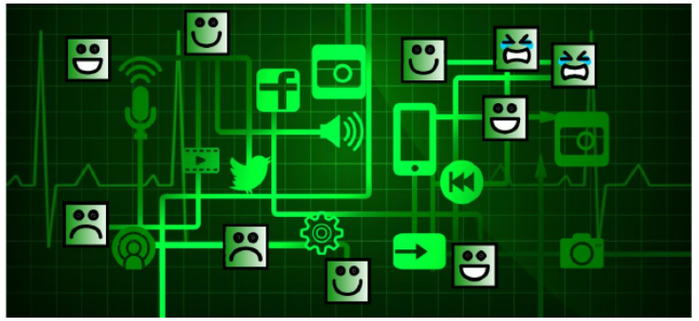
2027: Computational Modelling of Social Systems

- Seminar with exercises: 6 ECTS (2 + 2 SWS)
- To be taught in 2027 by David Garcia
- https://github.com/dgarcia-eu/CMSS-Konstanz
- Lectures and exercises on Agent-Based Modeling and Complexity Science
- Topics:
  - Modeling segregation: Schelling's model
  - Modeling cultures: Axelrod's model
  - Spreading of behavior
  - Opinion dynamics
  - Network growth models
  - Social resilience
- Python exercises with Mesa



Path to Master Thesis

- **Recommendation:** take *Colloquium in Social Data Science* the semester before your master thesis
- Preparation by joining group research updates (Tuesdays 13:30-15:00)
- Students prepare their thesis plan during the semester and present it for input at the end
- Manual registration by me – contact me before to discuss coursework and potential topics
- **Second recommendation:** take my group's courses and read our papers



Summary

- **Computational Social Science**
  - Using new sources of data and computational methods in the social sciences: the three meanings of the word *computational*
- **Digital Behavioral Data in the Social Sciences**
  - Social media, search, mobile phones… Risks: representation problems
- **New Methods in the Social Sciences**
  - Text analysis as an example. Risks: privacy
- **Computational Analysis of Social Science Theory**
  - Agent-Based Modeling to analyze models of social systems

Next sessions: Q&A and exam review